



Semester 2 Examinations, 2015/2016

Exam Code(s)	3BME1, 3BMS2, 3BS9, 4MBE2, 4BMS2, 4BS2, 1HA1
Exam(s)	3rd Arts, 3rd Science, 4th Arts, 4th Science
Module Codes	MA378, MA531
Module	Numerical Analysis II
Paper No	1
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Internal Examiner(s)	Prof. Graham Ellis Dr N. Madden ★

Instructions: **Answer THREE questions**

Duration 2 Hours
Number of Pages 2 (including this page)
Discipline) Mathematics

Requirements:

Release in Exam Venue	Yes ☒	No ☐
Release to Library	Yes ☒	No ☐
Statistical/ Log Tables	Yes ☒	No ☐
Graph Paper	None	
Log Graph Paper	None	
Graphic material in colour	Yes ☐	No ☒

- Q1. (a) [11 MARKS] State the *Polynomial Interpolation Problem*, and explain how it can be solved using Lagrange Polynomials.
- (b) [11 MARKS] State and prove Cauchy's theorem for $f(x) - p_n(x)$.
- (c) [11 MARKS] Let p_3 be the polynomial that interpolates $f(x) = \sin(x/2)$ at the points $x_0 = 0$, $x_1 = 1/3$, $x_2 = 2/3$ and $x_4 = 1$. Use Cauchy's Theorem to give a bound for the error at $x = 1/2$. (You don't have to give a formula for p_3).

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- Q2. (a) [11 MARKS] Define *piecewise linear* interpolation.
Denote by l , the piecewise linear interpolant to f on $n+1$ *equally spaced points* $[a, b]$. Let $h = (b-a)/n$. Use Cauchy's theorem to give an upper bound for $\max_{a \leq x \leq b} |f(x) - l(x)|$, in terms of h and f'' .
- (b) [11 MARKS] Let $f(x) = \log(x)$, $a = 1$, $b = 2$ and $n = 2$. Write down a formula for its piecewise linear interpolant l .
Give an upper bound for the maximum error on $[1, 2]$.
What value of n would you have to choose to ensure that $\max_{1 \leq x \leq 2} |f(x) - l(x)| \leq 10^{-4}$?
- (c) [11 MARKS] Let $\hat{l}$ be *any* piecewise linear approximation to a given function f on a fixed set of interpolation points, and l be the piecewise linear interpolant to f . Show that

$$\max_{x_0 \leq x \leq x_n} |f(x) - l(x)| \leq 2 \max_{x_0 \leq x \leq x_n} |f(x) - \hat{l}(x)|.$$

What is the significance of this result?

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- Q3. (a) [11 MARKS] Explain *Newton-Cotes Quadrature*.
Show how to derive the 3-point Newton-Cotes rule (i.e., Simpson's Rule).
- (b) [11 MARKS] What is meant by the *precision* of a quadrature method?
Show that Simpson's rule has precision 3.
- (c) [11 MARKS] Let $\{\hat{p}_n(x)\}_{n=0}^{\infty}$ be the sequence of orthogonal monic polynomials with respect to the inner product

$$(f, g) := \int_a^b f(x)g(x)dx.$$

- (i) Prove that $\hat{p}_n(x)$ is orthogonal to *all* polynomials of degree less than n .
- (ii) Prove that the zeros of the $\hat{p}_n(x)$ are simple (i.e., distinct).
- (iii) Prove that the zeros of the $\hat{p}_n(x)$ are all in the interval $[a, b]$.

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- Q4. (a) [20 MARKS] Suppose you wanted to use the Galerkin finite element method with piecewise linear basis functions (i.e., "*hat*" functions) to find an approximate solution to

$$-u''(x) + u(x) = f(x) \quad \text{on} \quad (0, 1) \quad \text{and} \quad u(0) = u(1) = 0, \quad (1)$$

on the uniform mesh $\{x_0, x_1, \dots, x_{n-1}, x_n\}$.

Give a formula for the basis functions.

State the variational formulation of (1).

Explain why the system matrix, A is tridiagonal.

Derive the entries $a_{i,i-1}$, $a_{i,i}$ and $a_{i,i+1}$, for $i = 1, 2, \dots, n-1$.

- (b) [13 MARKS] State and prove Céas lemma.

Explain its relevance to the errors in the finite element method.